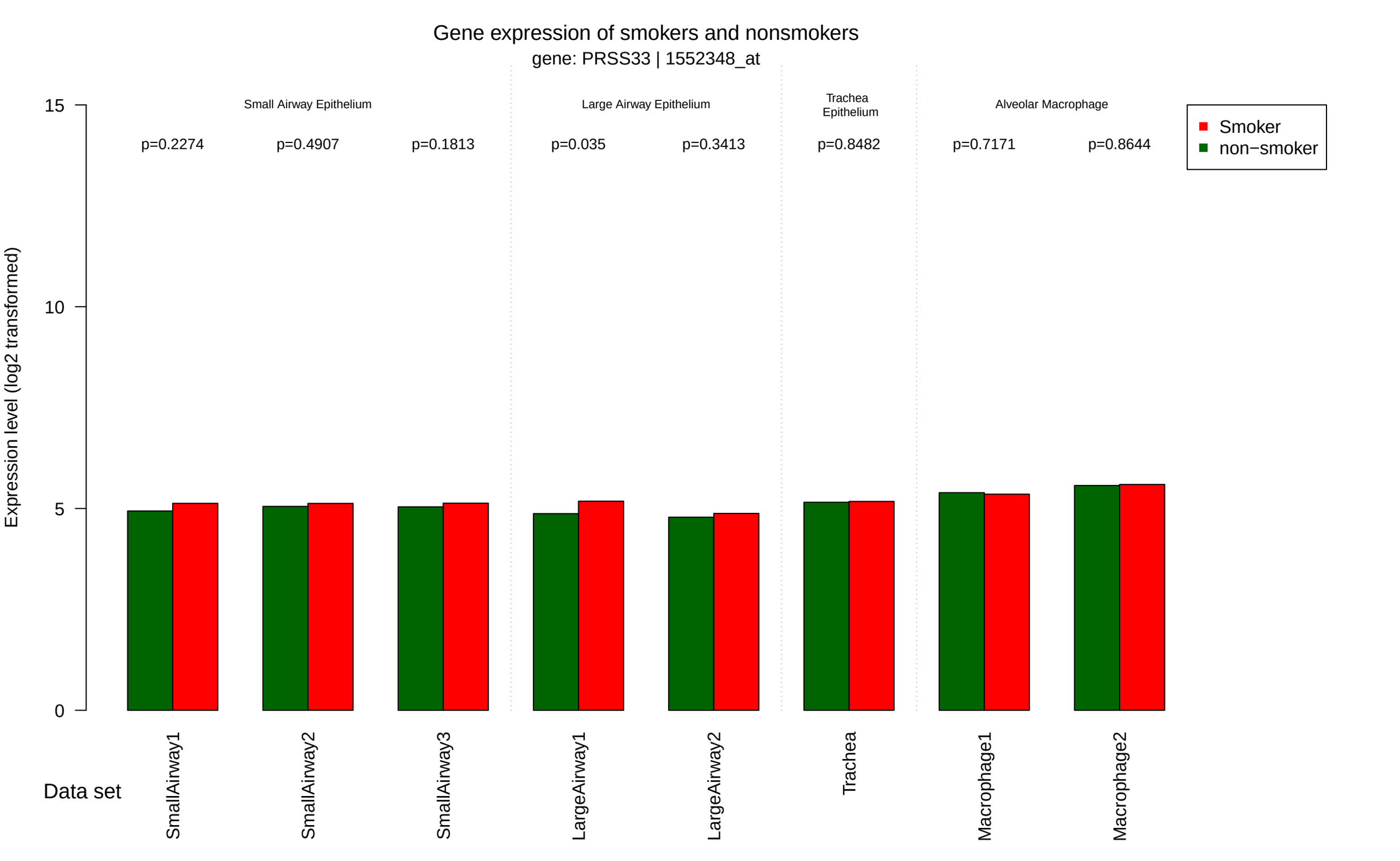
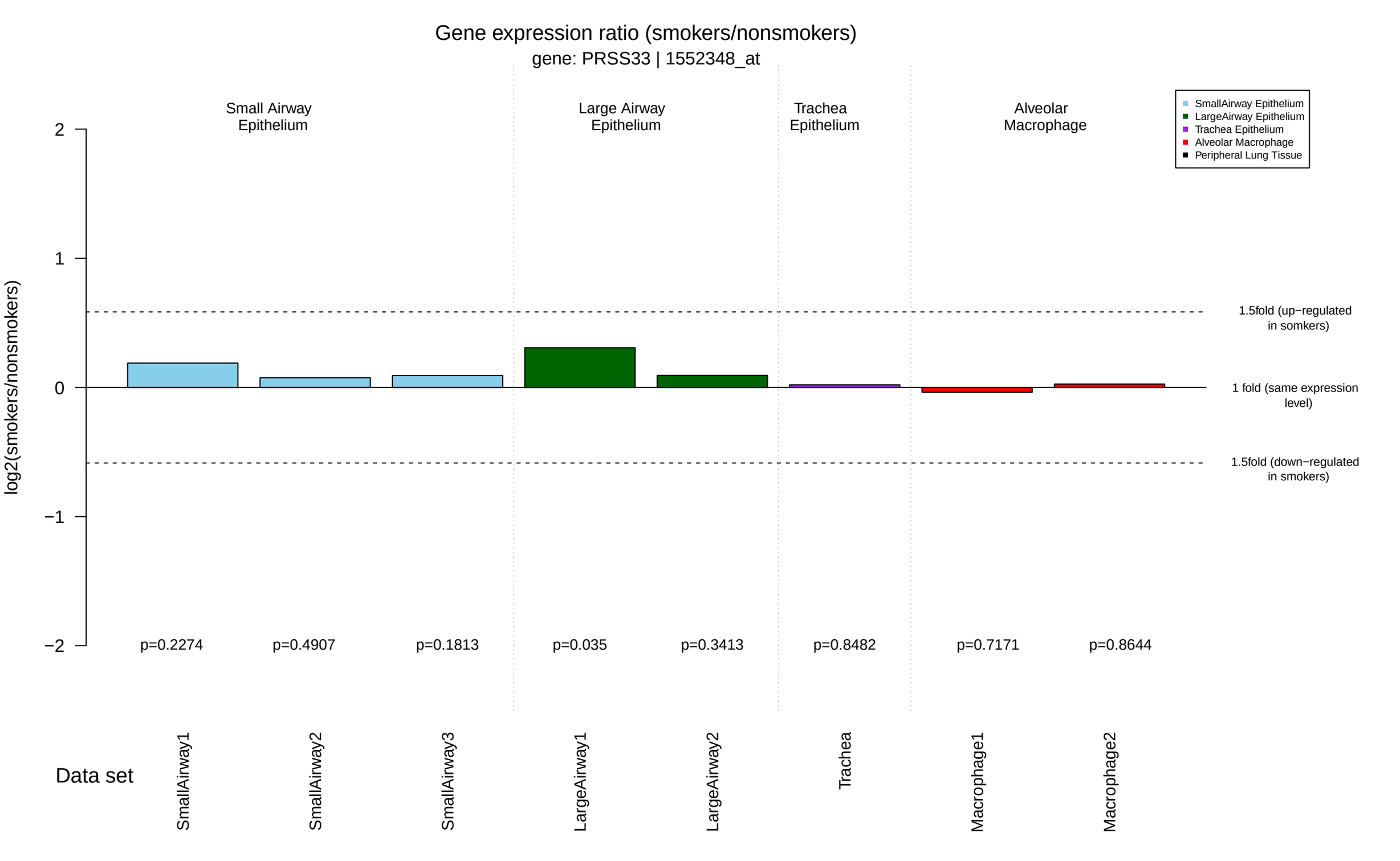
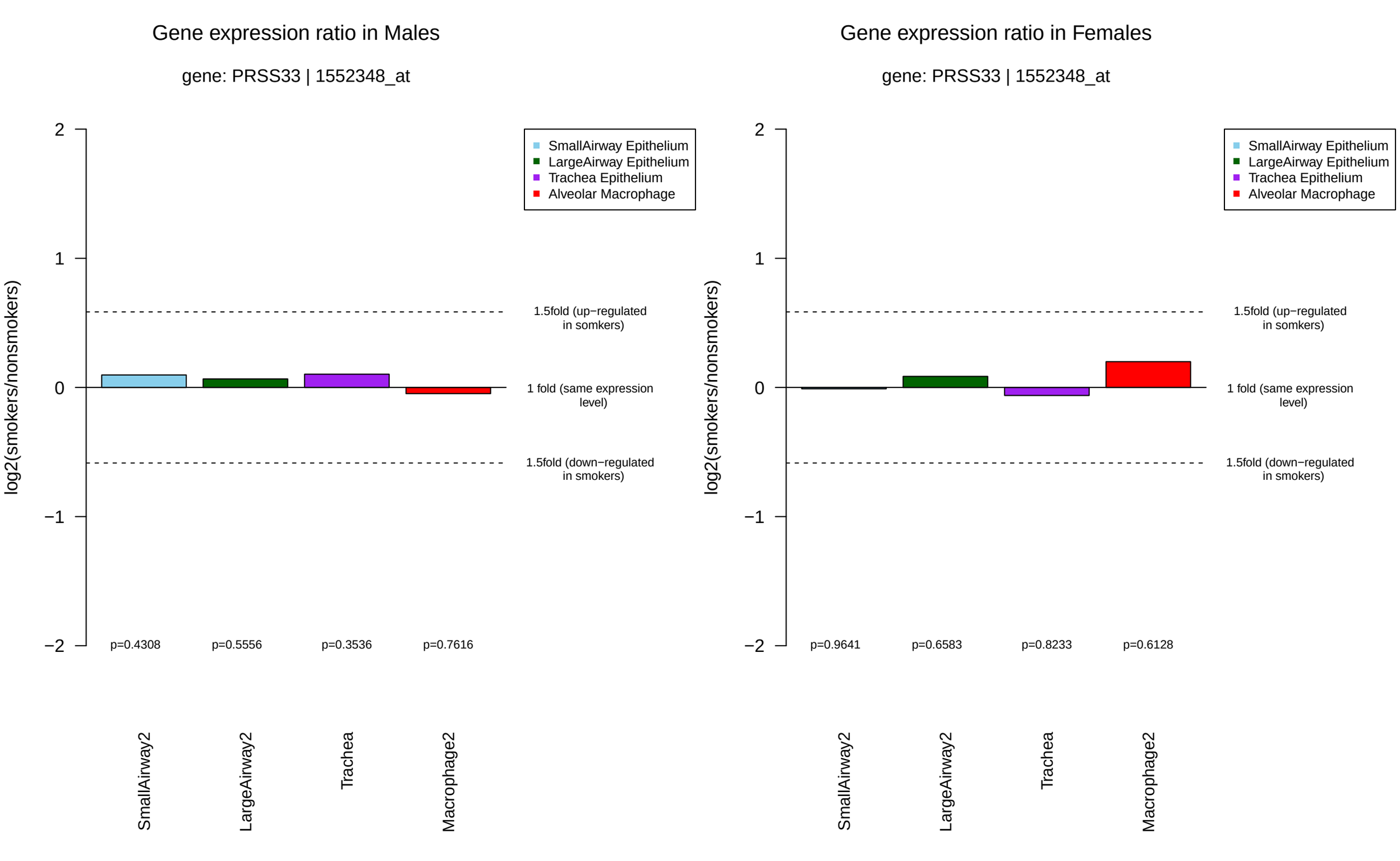
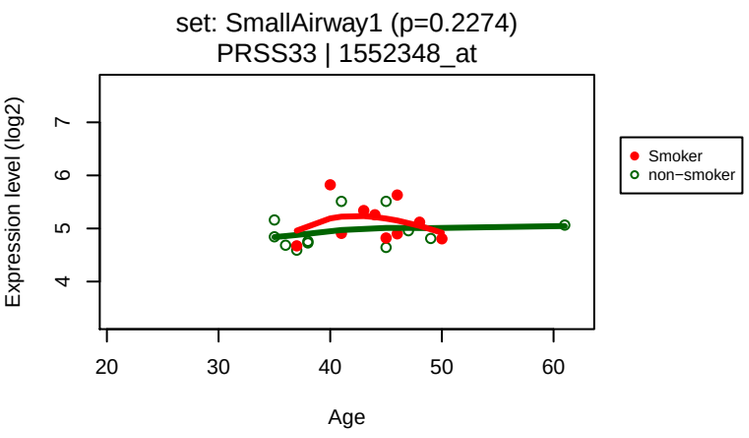
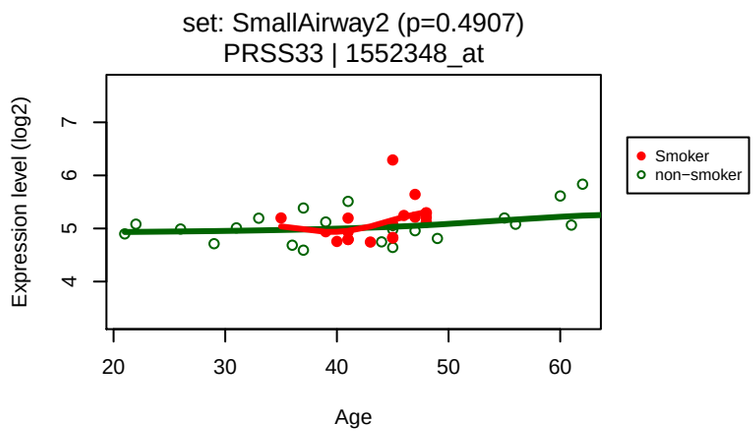
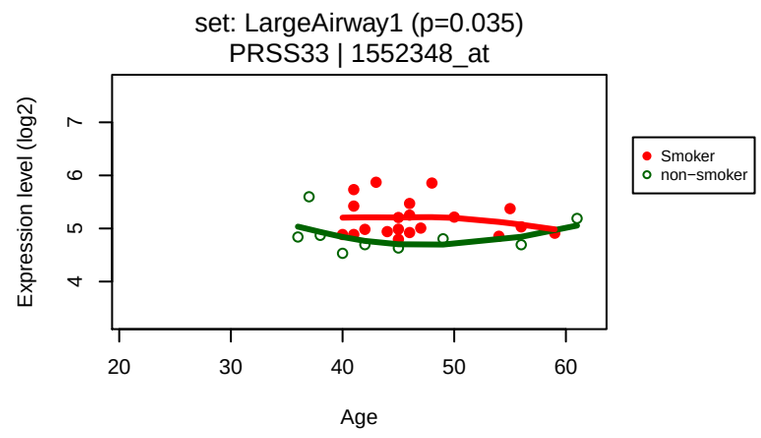

Gene expression of smokers and nonsmokers
gene: PRSS33 | 1552348_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2274
p=0.4907
p=0.1813
p=0.035
p=0.3413
p=0.8482
p=0.7171
p=0.8644
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

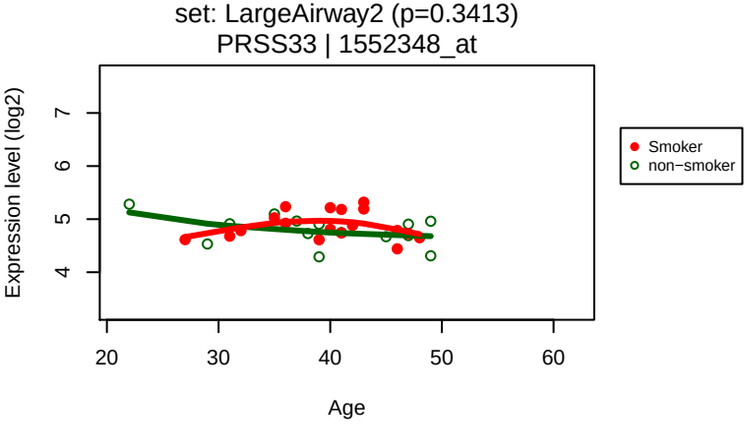

Gene expression ratio (smokers/nonsmokers)
gene: PRSS33 | 1552348_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2274
p=0.4907
p=0.1813
p=0.035
p=0.3413
p=0.8482
p=0.7171
p=0.8644
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

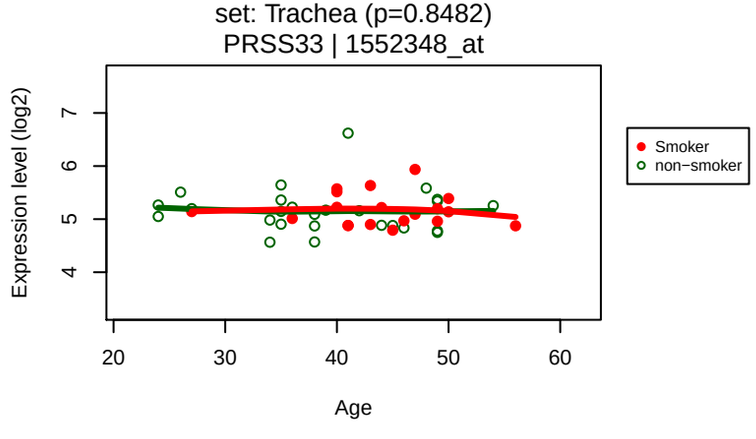

Gene expression ratio in Males
gene: PRSS33 | 1552348_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4308
p=0.5556
p=0.3536
p=0.7616
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PRSS33 | 1552348_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9641
p=0.6583
p=0.8233
p=0.6128
SmallAirway2
LargeAirway2
Trachea
Macrophage2

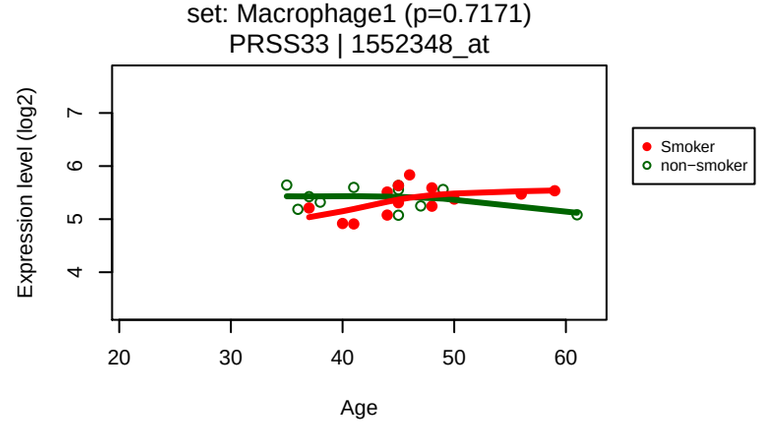

set: SmallAirway1 (p=0.2274)
PRSS33 | 1552348_at
Expression level (log2)
Age
Smoker
non−smoker

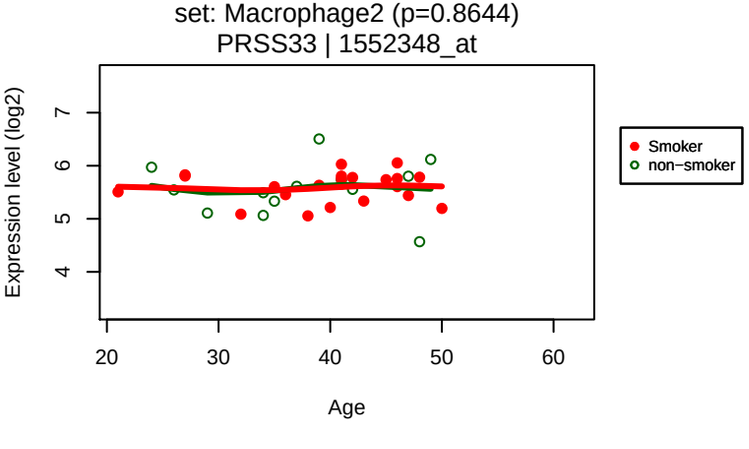

set: SmallAirway2 (p=0.4907)
PRSS33 | 1552348_at
Expression level (log2)
Age
Smoker
non−smoker

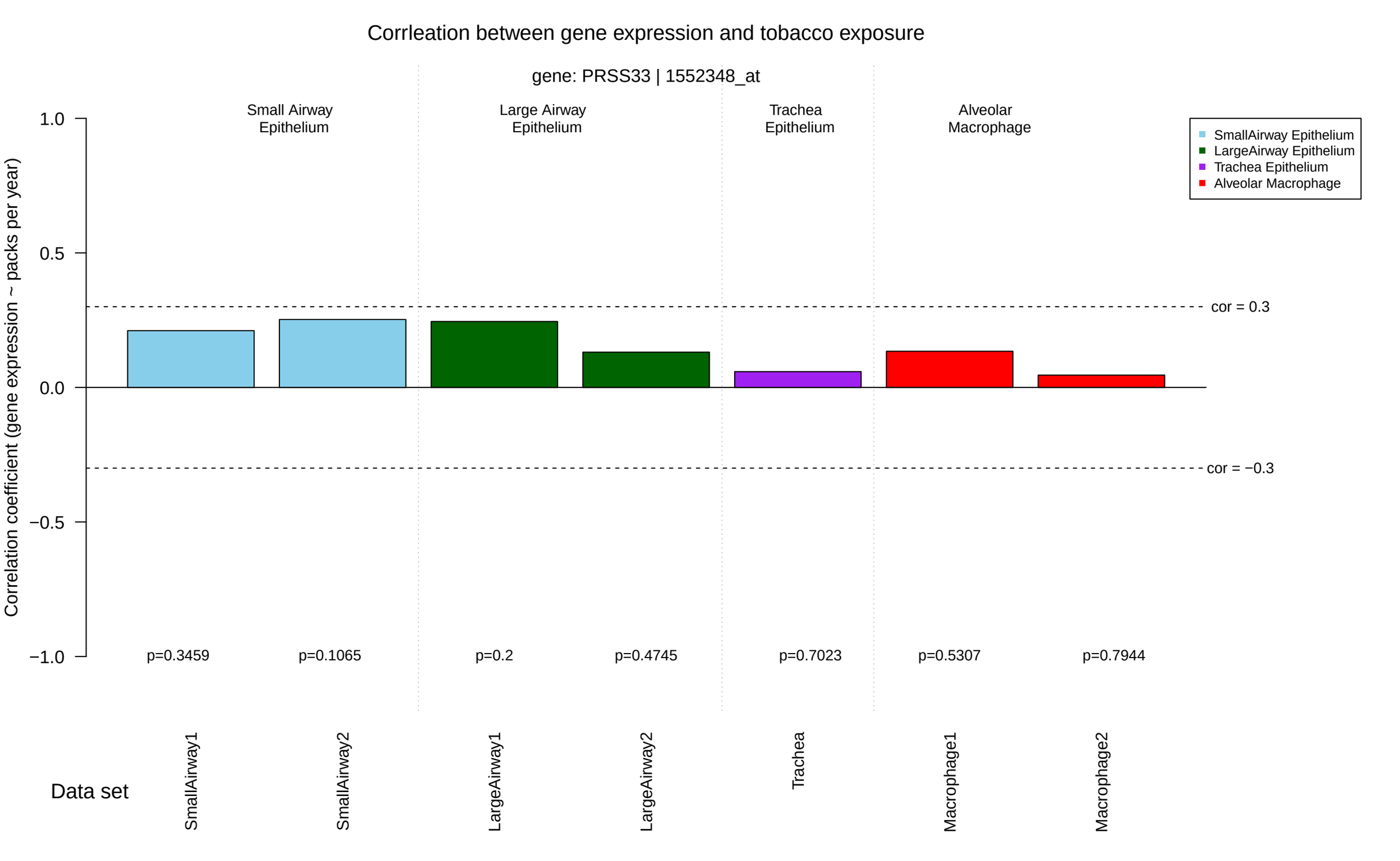

set: LargeAirway1 (p=0.035)
PRSS33 | 1552348_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3413)
PRSS33 | 1552348_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8482)
PRSS33 | 1552348_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7171)
PRSS33 | 1552348_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8644)
PRSS33 | 1552348_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PRSS33 | 1552348_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3459
p=0.1065
p=0.2
p=0.4745
p=0.7023
p=0.5307
p=0.7944
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2